

Cycle 3 Infographic Summary

Gender and Current Cigarette Use (YRBS 2007)

1 RESEARCH QUESTION



Is the proportion of current cigarette use different between male and female students?

2 GROUP DEFINITION



- Female students ($n = 6,740$)
- Male students ($n = 6,572$)

3 RESPONSE VARIABLE



"Current cigarette use" (binary)

- 1 = current cigarette use
- 0 = no current cigarette use

4 METHOD

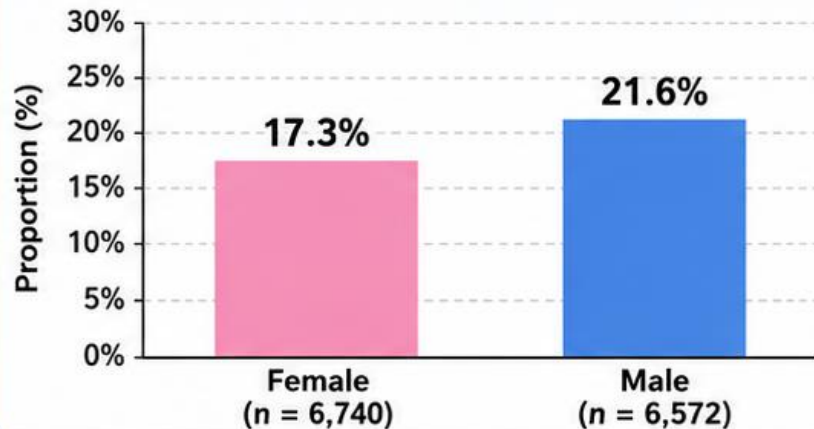


- Two-proportion z-test
- Significance level: $\alpha = 0.05$

5 KEY RESULTS

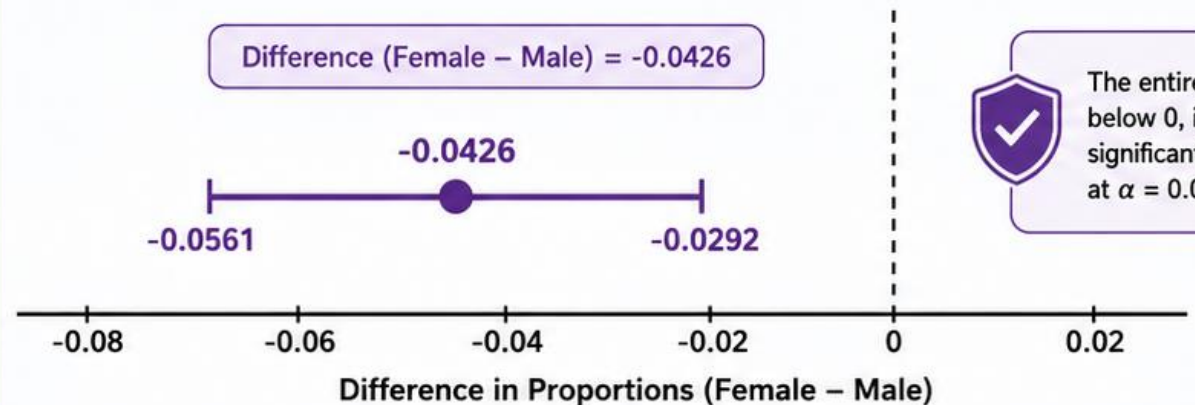
- ★ Female: $1,167 / 6,740 = 17.3\%$
- ★ Male: $1,418 / 6,572 = 21.6\%$
- ★ Difference (Female – Male) = -0.0426
- ★ 95% CI = $(-0.0561, -0.0292)$
- ★ $z = -6.2148$
- ★ $p < 0.001$

CURRENT CIGARETTE USE BY GENDER
(Proportion of Students)



↑
Male students have a higher proportion of current cigarette use (**21.6%**) than female students (**17.3%**).

95% CONFIDENCE INTERVAL FOR DIFFERENCE IN PROPORTIONS
(Female – Male)



6 CONCLUSION



Reject H_0 . The proportion of current cigarette use differs significantly between male and female students. Male students show a higher current cigarette use proportion than female students.

7 INTERPRETATION



This result shows an association in survey data and should not be interpreted as causation.